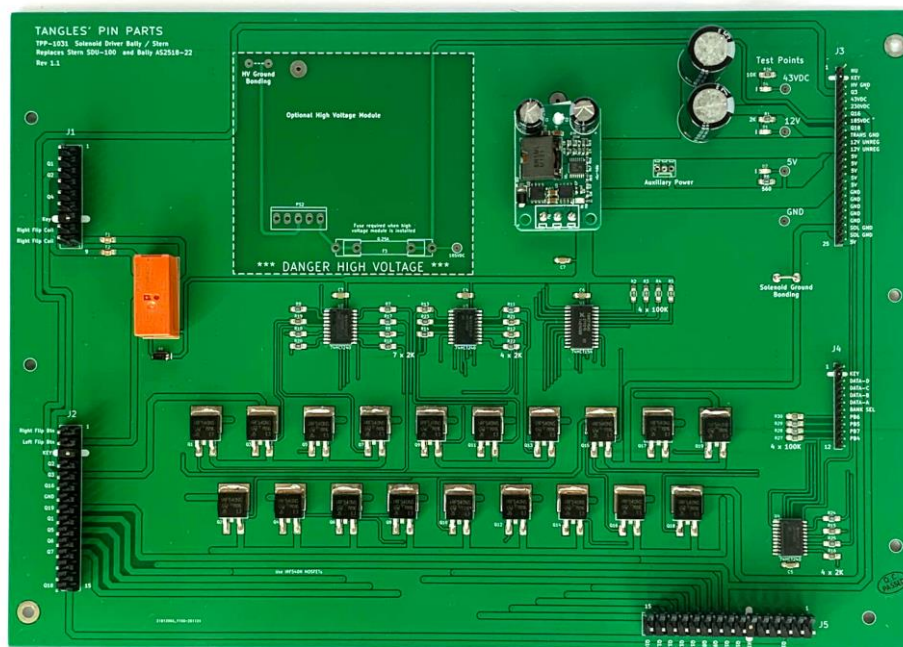


TANGLES' PIN PARTS

TPP-1031

Solenoid Driver For Bally / Stern Systems

Replacement for AS-2518-22 and SDU-100 boards



SUITABLE FOR GAMES WITH LED DISPLAYS

Description

The Tangles Solenoid Driver has been designed for Bally and Stern games with upgraded LED display modules. This board offers a significantly reduced complexity and enhanced performance over the original boards.

Running LED displays places a significant load on the game's 5V power supply. This board has been designed specifically for games that have upgraded to LED displays. It provides ample 5v power (up to 25 Watts) using a cool running switch-mode power supply.

Removing the High-Voltage circuitry reduces the board's complexity and increases the game's safety.

Features

- 25Watt Switched Mode Power Supply. Supplies up to 5A of 5 volt power with a better than 90% efficiency.
- Super high quality gold contacts on the mini Molex pins make for a reliable connection to the wiring harness.
- A direct replacement for the original board... but for LED display systems only!
- Additional fuse protection for your flipper circuits. No more burnt traces on the PCB for these circuits.
- No soldering required.
- High Efficiency means your transformer, rectifier and 5V regulators generate significantly less heat.
- No High Voltage on the PCB! This feature limits this board to LED display systems only, but significantly improves the safety of the system.
- MOSFETs installed make a snappy fast action on all coils.
- Highest Quality Japanese capacitors used throughout.
- Uses top tier parts sourced through reliable supply chains. This board utilises modern components and manufacturing to extend the life and reliability of your Pinball machine.
- Over-current and short circuit protection provided. This 5V power circuitry is quite bullet proof. It safely shuts down when a fault is detected without even blowing a fuse!

*****NOTE*** This board is sold without a HV power supply; hence is NOT compatible with High Voltage (original) display modules.**

Installation

Warning: High voltages are present inside the pinball machine. Turn off the power and unplug the mains before you commence work.

1. Ensure power is off.
2. Open the back cabinet and unplug the 5 Molex connectors.
3. Remove the 2 securing screws (Bottom left and top right)
4. Unclip the 4 mounting posts and remove the old board.
5. Install the TPP-1031 onto the 4 mounting posts (push on)
6. Secure the two mounting screws.
7. Reconnect the 5 Molex connectors taking care to align the "Key" with the missing pin on plug.

Game Compatibility

The **Tangles TPP-1031 Solenoid Driver Board** is compatible with the following games when they are fitted with LED displays:

STERN

Ali, Big Game, Catacomb, Cheetah, Cosmic Princess, Dracula, Dragonfist, Flight 2000, Freefall, Galaxy, Iron Maiden, Lazer Lord, Lectrnamo, Lightning, Magic, Memory Lane, Meteor, Nine Ball, Nugent, Orbitor 1, Pinball, Quicksilver, Seawitch, Split Second, Star Gazer, Stars, Stingray, Trident, Viper, Wild Fyre

BALLY

Black Jack, Black Pyramid, BMX, Centaur, Centaur II, Cybernaut, Dolly Parton, Eight Ball, Eight Ball Deluxe, Eight Ball Deluxe Ltd, Elektra, Embryon Mystic, Evil Knievel, Fathom, Fireball Classic, Fireball II, Flash Gordon, Freedom, Frontier, Future Spa, Grand Slam, Harlem Globe Trotters, Hotdoggin, Kings of Steel, Kiss, Lost World, Mata Hari, Medusa, Mr. and Mrs. Pacman, Night Rider Vector, Nitro Ground Shaker, Paragon, Playboy, Power Play, Rapid Fire, Rolling Stone, Silver Ball Mania, Six Million Dollar Man, Skate Ball, Space Invaders, Speakeasy, Spectrum, Spy Hunter, Star Trek, Strikes and Spares, Super Sonic, Viking, Voltan, Xenon, X's and O's

Trouble Shooting

Problem	Comments
No Lights / Power	• Check fuses on the Rectifier board • Check the 43V and 12V LEDs are lighting on the TPP-1031. These indicate power getting to the board.
One or more solenoids not working	• Check the appropriate Molex connection for the solenoid. Consult your game manual to identify the "Q" number. Replace the Molex pin in the wiring harness. • A missing or open circuit flyback diode (in the back of each coil) will result in damage to the driving MOSFET. • Consult the MOSFET test and replace section.
12V LED is lit, but 5v LED is not.	• There is a short circuit on the 5v line.

Testing

Follow the instruction from your game manual for LAMP testing.
This is typically done by:

- TURN ON POWER
- PRESS SELF TEST BUTTON ON INSIDE OF FRONT DOOR TO ENTER TEST MODE
- PRESS THE SELF TEST BUTTON TWO MORE TIMES TO ENTER THE SOLENOID TEST SEQUENCE. EACH SOLENOID WILL BE ENEGISED ONE AT A TIME IN A CONTINUOUS SEQUENCE.
- TURN POWER OFF THEN ON. GAME READY TO PLAY

Support and Other Resources

In the event that you require support for your product or assistance with installation please contact Tangles directly at tangles@tanglestech.com

Video illustrating the installation:

<http://bit.ly/TPP1031VID>

Updates to this manual will be posted here:

<http://bit.ly/TPP1031>